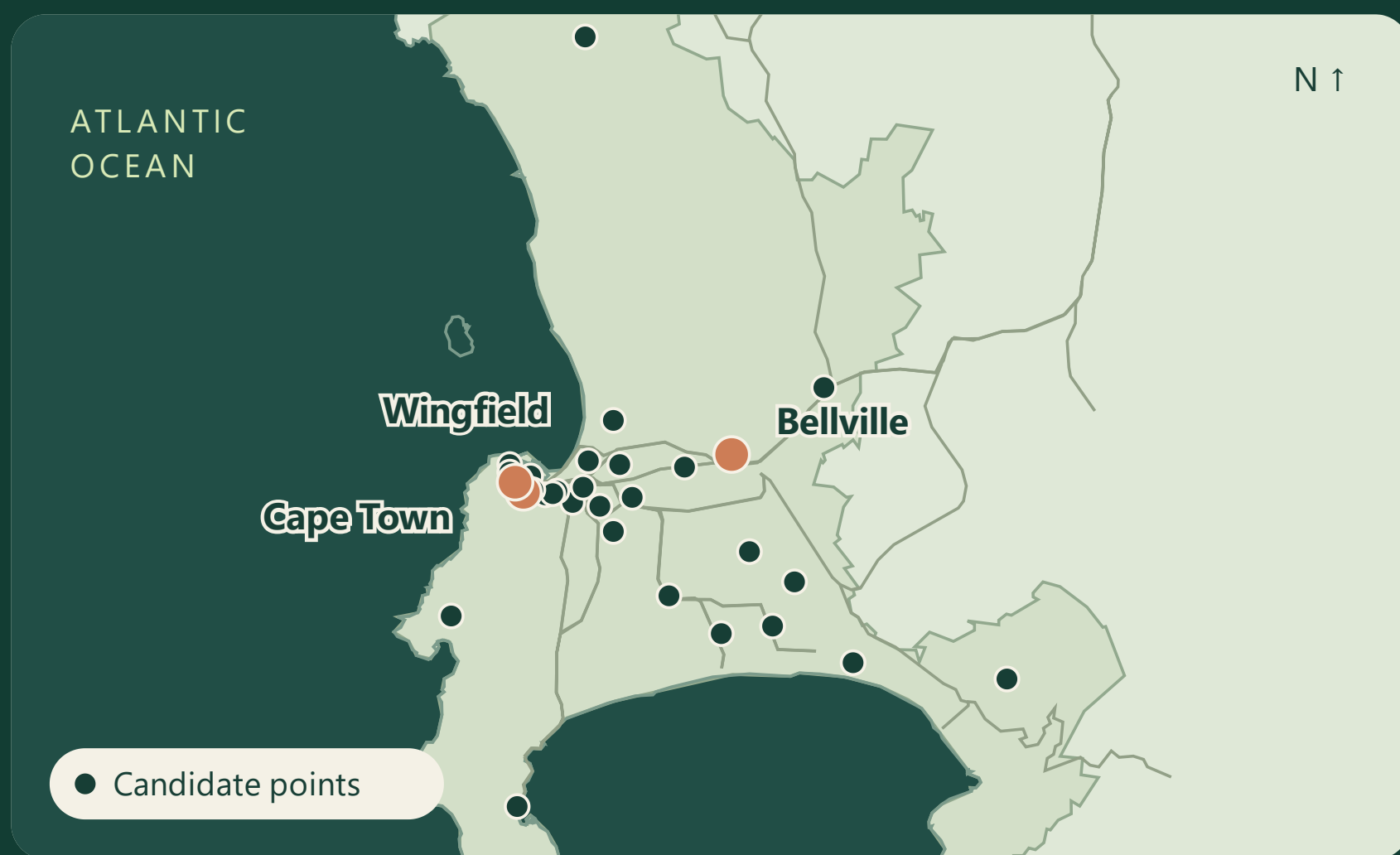


From data. To clearer decisions.

Python, GIS and visual storytelling
in one housing opportunity atlas.



30

candidate precincts

14

screening criteria

2

ranking lenses

Desktop screening study. Map context: © OpenStreetMap contributors.

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Data. Place. Clear explanation.

A connected approach to analytical work.

01 / ANALYSE

Python + data

Repeatable scoring,
scenarios and checks.

02 / LOCATE

GIS + context

Spatial relationships,
proximity and maps.

03 / MODEL

Power BI

Star schema and
DAX measures.

04 / EXPLAIN

Visual reporting

Interactive views,
reports and ebooks.

Demonstrated through the
Cape Town Housing Opportunity Atlas.

A point is only the start.

Add the surrounding context
to make location data useful.

14 / 30

candidate points within 1 km
of a mapped rail point

131

mapped rail points

1,642

mapped facility features

A reproducible proximity measure.
Walking routes and operating services
need their own verification.

Straight-line distance. OSM snapshot: 30 August 2026.
Facility features can include overlapping campus records.

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One city. Two lenses.

How priorities change a model’s shortlist.

CANDIDATE	EFFICIENCY	REDRESS
Culemborg	18	10
Helen Bowden	15	8
Wingfield	8	4

Make the assumptions visible.
Then discuss the decision.

Selected model ranks; 1 is highest. Efficiency and spatial redress lenses.

How stable is the shortlist?

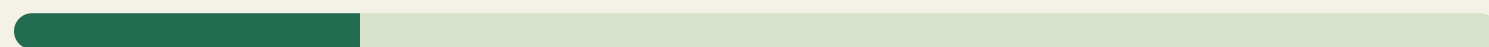
Test the result when assumptions move.

7

of 30

Stable under input changes

2,000 simulation draws



9

of 30

Stable under weight changes

5,000 simulation draws



Stable = top ten in at least 90% of draws.
A test of this model's consistency.

Inputs perturbed $\pm 15\%$; weights $\pm 30\%$ and renormalised.
These simulations do not estimate development success.

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One workflow. Many useful outputs.

Connect the calculation to the formats
people need to use.

SOURCE TABLES + GEOGRAPHIC CONTEXT



Python analysis workflow

Scoring · scenarios · sensitivity · checks

Web atlas

Maps + comparisons

Power BI model

Schema + measures

Publication

PDF + EPUB + web

Data preparation → derived measures
→ explainable analytical outputs

Portfolio outputs generated from the project's tables and spatial context.

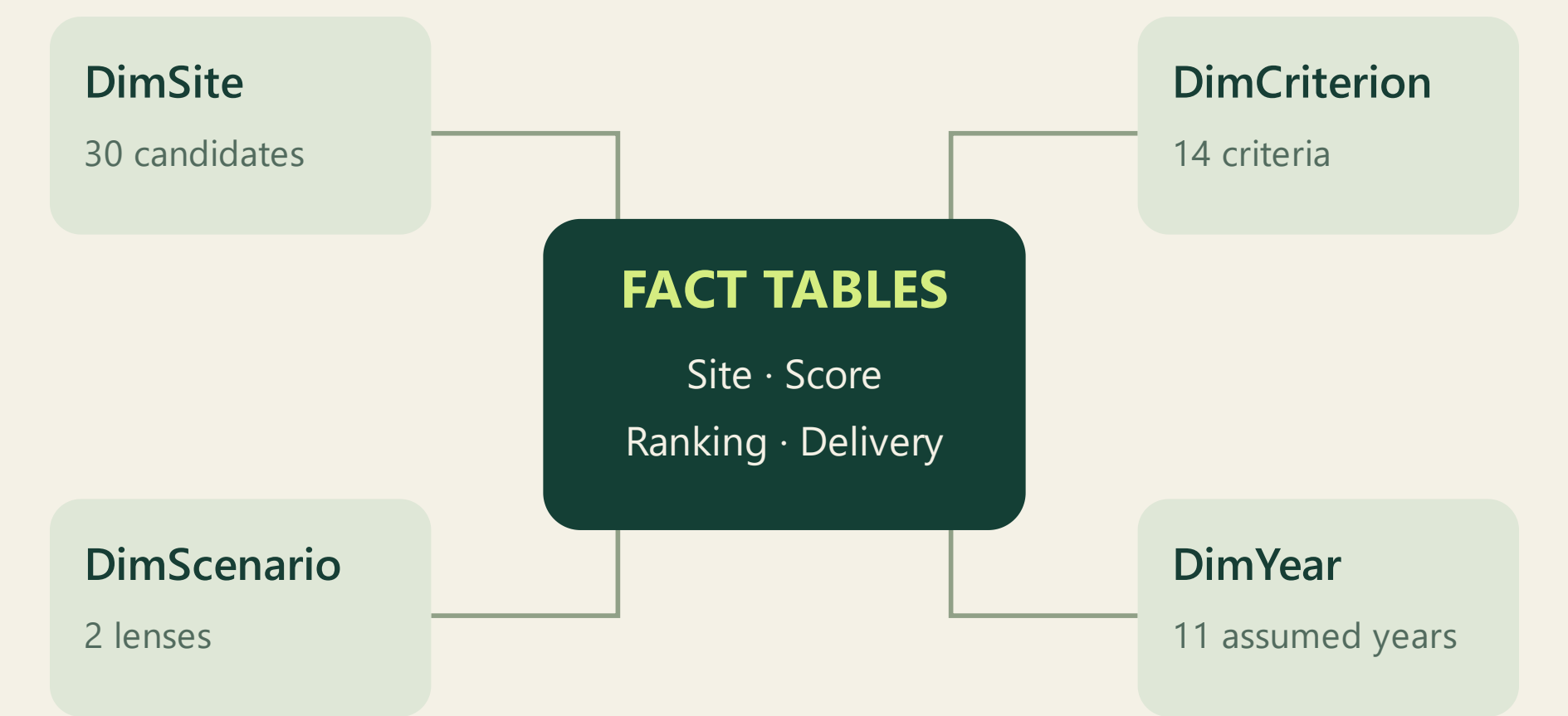
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Good reports start with a good model.

A star schema that makes analytical relationships explicit.



420

site × criterion records

60

site × scenario records

Conceptual schema view; source-defined model and DAX.
Delivery years are assumptions. Production deployment is not claimed.

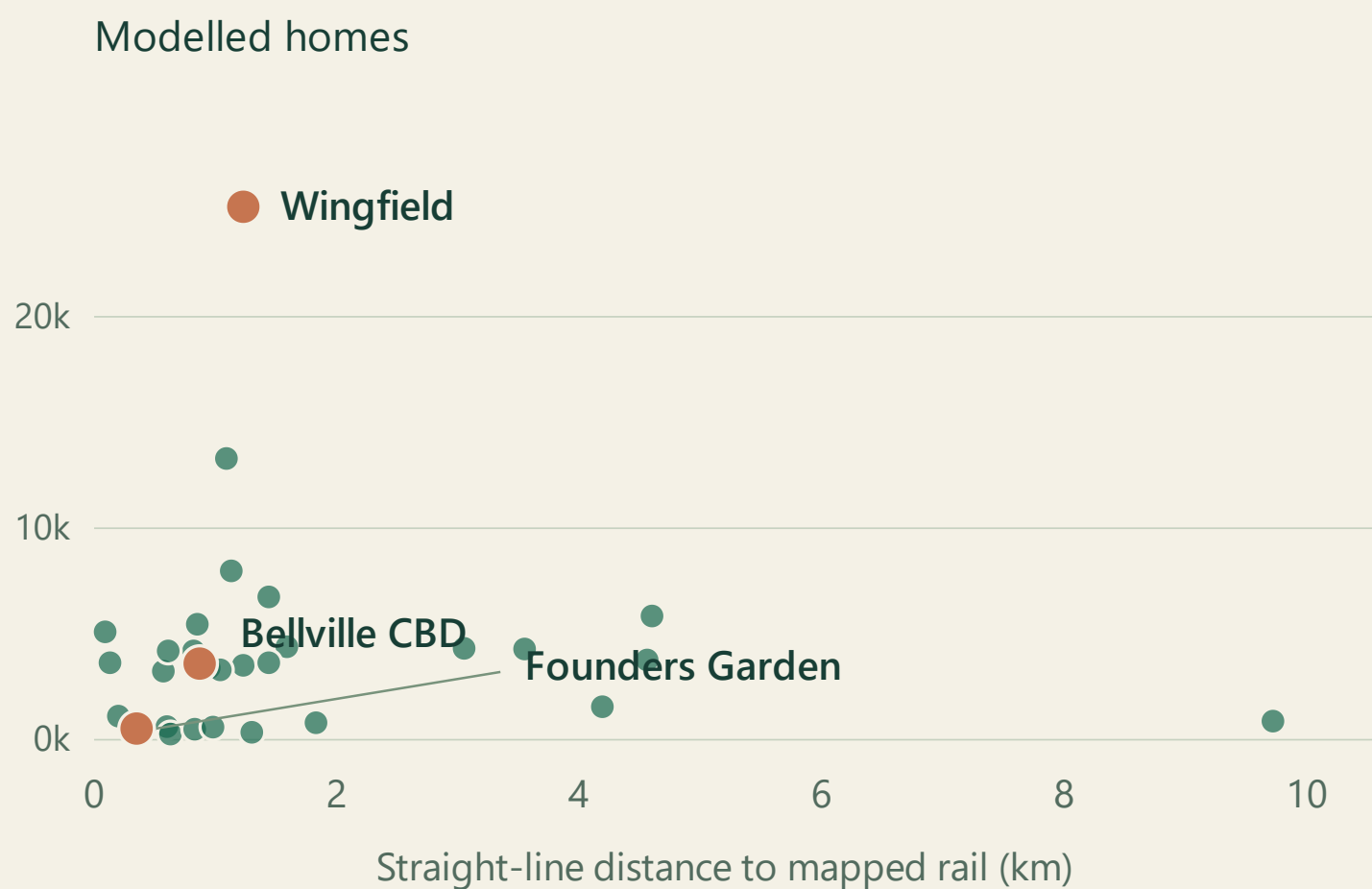
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Show the trade-off.

Compare modelled housing capacity with proximity to mapped rail.



Use the chart to find the next question.
Use the evidence to investigate it.

30 candidate points. Capacity uses unverified land and density inputs. Modelled homes are not approved or committed housing delivery.

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Give the analysis a readable form.

An illustrated report with an index,
site profiles and a clear evidence trail.



30

report pages

PDF

EPUB

Web edition

Executive summary · maps · methodology
Candidate directory · source notes · indexes

Schematic cover treatment of the project publication; PDF includes
bookmarks and internal navigation.

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Make the work easy to inspect.

Useful analysis has an evidence trail.

33 Python tests passed
for this project version

Trace the inputs

Source, calculation and assumption.

Check the experience

Browser, mobile and publication review.

Expose the gaps

Desktop estimates stay visible.

Tests check the implementation. Real-world feasibility and municipal records require independent verification.

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Open to remote work.

Roles, freelance engagements
and contract projects.



Python analytics
Spatial analysis + maps
Power BI modelling
Visual reports + publications

Have a relevant opportunity?
Message me on LinkedIn.

Let's make your data more useful.

Bring me the question.

Let's shape the analysis and the output.

01 Understand the data

Structure, sources and evidence gaps.

02 Explore the decision

Spatial context, scenarios and comparisons.

03 Explain the result

Interactive views and readable reports.

Explore the atlas. Read the report.

Let's talk about your project.

Portfolio case study available on request.

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Rent is only part of the cost.

Illustrative monthly budget
on R20,000 gross household income.

Rent	R6,000
Utilities	R1,000
Transport	R1,800

44% of gross income
for the three costs combined

A well-located home also needs
an achievable household budget.

Illustration, not survey data. Tax, food, care and debt are excluded.
These amounts are assumptions, not a rental forecast.

Rent controls: design matters.

Evaluate who gains protection,
who can find a home, and what gets built.

Protect tenants

Predictable costs and secure occupancy.

Keep homes available

Maintenance, investment and rental supply.

Measure both

Existing tenants and people seeking a home.

Test coverage, cost-review rules,
enforcement and supply effects.
Pair protection with affordable homes.

Policy design questions, not adopted Cape Town rules.
Evidence and legal context: report pages 28 and 30.

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Affordable homes need a programme.

Combine land, services, finance and tenant protection.

01 Unlock suitable land

Verify rights, services and delivery milestones.

02 Fund lasting affordability

Affordable rental, operating costs and upkeep.

03 Support households

Targeted help and accessible dispute resolution.

04 Track occupied homes

Costs, access, conditions and displacement.

Proposed programme for appraisal, not an adopted or funded plan.
Read the six-page policy section in the updated report.

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